

PERT/CPM & Crashing Report

Schedule analysis, completion probability and crash-versus-damages decision

Deliverable 1 of 6 · Case Study no. 39 · PortFlow

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Case	PortFlow - Container Terminal Yard and Gate Operations		

1. Objective and Method

This report finds the critical path, float, completion probability and the financially justified crashing decision for the PortFlow project, using only the figures given in the case study. The contract deadline is **34 weeks**, with liquidated damages of **₹12 lakh per week** of delay.

PERT turns three estimates per activity (O optimistic, M most likely, P pessimistic) into an expected time and a measure of uncertainty. **CPM** then uses the expected times to find the longest path (critical path) and the float of each activity.

$$TE = (O + 4M + P) / 6 \quad | \quad \sigma = (P - O) / 6 \quad | \quad \text{Variance} = \sigma^2$$

2. Activity Estimates

Activity	Predecessor	O	M	P	TE	σ	Variance
A	-	4	6	8	6	0.667	0.444
B	A	3	6	15	7	2.000	4.000
C	A	2	4	6	4	0.667	0.444
D	B	6	8	10	8	0.667	0.444
E	C	3	5	7	5	0.667	0.444
F	D, E	4	6	14	7	1.667	2.778
G	F	2	3	4	3	0.333	0.111

Worked example: $B = (3 + 4 \times 6 + 15) / 6 = 42 / 6 = 7$ weeks; $\sigma = (15 - 3) / 6 = 2.0$.

3. Network Diagram

Activity-on-node network. The critical path (red) is A → B → D → F → G.

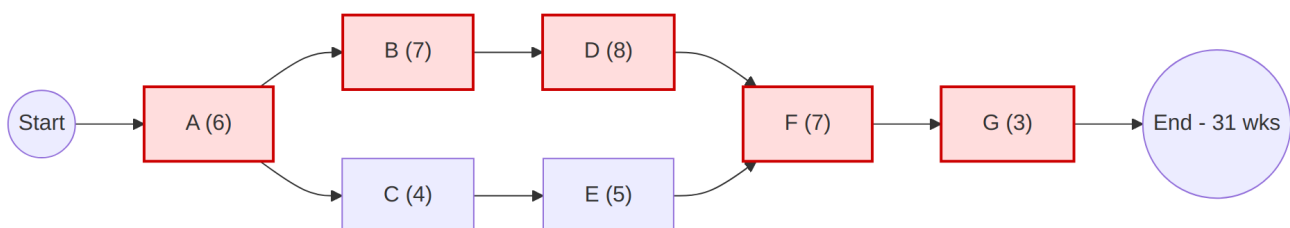


Figure 1: PERT network with expected durations (weeks); critical path in red

4. Path Analysis and Critical Path

Path	Calculation	Length
A-B-D-F-G	6 + 7 + 8 + 7 + 3	31 weeks (critical)
A-C-E-F-G	6 + 4 + 5 + 7 + 3	25 weeks

Critical path: A → B → D → F → G. Expected project duration: 31 weeks. F needs both D and E to finish, so the longer route decides the finish date.

5. Float Analysis

ES and EF come from the forward pass; LS and LF from the backward pass. Total float = LS – ES.

Act	TE	ES	EF	LS	LF	Total float	Critical?
A	6	0	6	0	6	0	Yes
B	7	6	13	6	13	0	Yes
C	4	6	10	12	16	6	No
D	8	13	21	13	21	0	Yes
E	5	10	15	16	21	6	No
F	7	21	28	21	28	0	Yes
G	3	28	31	28	31	0	Yes

Where slack genuinely exists: only on C and E, and the 6 weeks is **shared** by the path, not owned by each activity. C has free float 0 (if C slips, E's start moves); E has free float 6. If C uses 4 weeks, E has only 2 left. Every other activity has zero float, so any slip there delays the project.

6. Project Variance and Probability of Finishing within 34 Weeks

Only critical-path variances add: $0.444 + 4.000 + 0.444 + 2.778 + 0.111 = \mathbf{7.778}$, so the project $\sigma = \sqrt{7.778} = \mathbf{2.789 \text{ weeks}}$. Activities B and F contribute 6.78 of the 7.78, so they are the main source of schedule uncertainty.

$$Z = (\text{Deadline} - \text{Expected}) / \sigma = (34 - 31) / 2.789 = \mathbf{1.076} \rightarrow P(\text{finish} \leq 34 \text{ weeks}) \approx \mathbf{86\%}$$

Target (weeks)	Z	P(on time)
31 (expected)	0.00	50%
34 (deadline)	1.08	≈ 86%
37	2.15	≈ 98.4%
39	2.87	≈ 99.8%

This uses the normal approximation, the standard PERT assumption. There is therefore about a 14% chance of missing the deadline.

7. Crash-versus-Damages Financial Analysis

7.1 Crash options (given data)

Activity	Max crash	Cost / week	vs penalty ₹12 L	Worth crashing?
B	3 weeks	₹7 L	Cheaper	Yes (first)
F	2 weeks	₹9 L	Cheaper	Yes (second)
D	2 weeks	₹15 L	Costlier	Never

All three activities are on the critical path. **Decision rule:** crash a week only if its cost is less than the ₹12 lakh penalty it avoids.

7.2 Baseline test: is any crashing needed now?

The expected finish is 31 weeks, already 3 weeks before the deadline, so expected weeks late = 0. Allowing for uncertainty, the expected overrun is $\sigma[\phi(z) - z(1 - \Phi(z))]$ ≈ 0.20 weeks, an expected penalty of about **₹2.4 lakh**. Crashing B by 3 weeks would cost **₹21 lakh**.

Baseline decision: do not crash. Spending ₹21 L to avoid an expected ₹2.4 L penalty destroys value.

7.3 Week-by-week comparison if the forecast finish exceeds 34 weeks

Weeks late	Penalty if not crashed	Cheapest crash cost	Net saving
1	₹12 L	$B \times 1 = ₹7 L$	+₹5 L
2	₹24 L	$B \times 2 = ₹14 L$	+₹10 L
3	₹36 L	$B \times 3 = ₹21 L$	+₹15 L
4	₹48 L	$B \times 3 + F \times 1 = ₹30 L$	+₹18 L
5	₹60 L	$B \times 3 + F \times 2 = ₹39 L$	+₹21 L
6+	₹72 L+	Only D left (₹15 L > ₹12 L)	Do not crash D

Crashing B and F does not change the critical path: with B crashed to 4 weeks the main path is 28 weeks, still longer than the 25-week path. The maximum worthwhile crashing is therefore 5 weeks (B 3 + F 2), costing ₹39 L against ₹60 L of penalty, a saving of ₹21 L.

8. Conclusion and Recommendation

1. Critical path A-B-D-F-G, expected duration 31 weeks, project $\sigma = 2.79$ weeks.
2. $P(\text{finish within 34 weeks}) \approx 86\%$. Float (6 weeks, shared) exists only on C and E.
3. **Do not crash at baseline.** Keep a pre-approved trigger: if the forecast finish exceeds 34 weeks, crash **B** (up to 3 weeks, ₹21 L) first, then **F** (up to 2 weeks, ₹18 L). **Never crash D.**
4. The ₹21 L reserve covers outcomes up to 37 weeks, about 98.4% of cases.